



AI TEAM TRAINING '27

TASK 4

Task 4: Implementing a Neural Network

The Scenario:

The World Cup is here, and this year the trophy won't be decided on the pitch alone - it will be decided in the analysis room. Our rivals, "Team Eigen," have deployed an AI-powered match-analysis system, but it's a disaster. Their model is unreliable, constantly misreading jersey numbers from broadcast footage - a '7' for a '1', a '3' for an '8'. Their analysts are feeding coaches heat maps and pass statistics for the wrong players, and their tactical calls are falling apart in real time. It's our chance to seize the advantage.

We need a system that's not just better, but flawless. A system that reads every jersey number on the pitch and gives our tactical staff a crystal-clear, real-time picture of the entire match - allowing them to make the split-second, match-winning substitutions and tactical switches we're famous for.

Your Mission:

Your mission is to build our next-generation vision system from the ground up. This isn't just about using off-the-shelf parts; it's about understanding the core mechanics to ensure our system is robust, accurate, and unbeatable. The campaign is divided into four critical phases:

1. Phase 1: Build the Playmaker. Every great side is built around the player who makes everything tick. You will start by building a custom automatic differentiation engine (the Value object) - the playmaker through which every gradient in your network will flow. Nothing works until this works.
2. Phase 2: Pick the Squad. With your playmaker ready, you will assemble the team: the Neuron (your individual players), the Layer (your lines - defense, midfield, attack), and the MLP (your full formation). A network like MLP(64, [32, 10]) is your team sheet.
3. Phase 3: Matchday. The group stage begins. You will train your squad to read handwritten-style jersey numbers (an 8x8 mini version of the famous MNIST dataset), then study its performance like a proper analyst - training curves, confusion matrices, and all - to prove your system outclasses Team Eigen's.
4. Phase 4: Going Pro. After proving your mastery by building a system from scratch, your final test is the professional league: re-implement the solution in PyTorch and unleash it on the real MNIST dataset - 70,000 images your scratch engine could never handle - demonstrating you're ready for any challenge.

Technical Requirements:

You are required to complete all TODO sections in the notebook, which will guide you through:

- Building a Value object with a complete backpropagation mechanism.
- Constructing the Neuron, Layer, and MLP classes from scratch.
- Implementing a training loop to train your custom model on the scikit-learn digits dataset (a mini 8x8 version of MNIST, sized so your pure-Python engine can handle it).
- Producing and interpreting the training loss curve and confusion matrix for your model - a working analyst watches the match, not just the final score.
- Rebuilding the model in PyTorch and training it on the real MNIST dataset (70,000 images at 28x28) to compare against your from-scratch implementation.
- Answering the three Understanding Check sections in your own words. These are graded and will be discussed in your evaluation session - we care about your reasoning at least as much as your code.

Setting Expectations:

Read this before you panic about your accuracy. With the baseline setup (MSE loss + tanh), your from-scratch model is expected to reach only around 50-60% test accuracy - that is the known ceiling of this configuration, not a bug in your code. Understanding why is part of the task. The bonus objectives below are what unlock 90%+, and they are where you can truly stand out. Also expect the from-scratch training to be slow (minutes per epoch) - that pain is intentional, and Phase 4 is the payoff.

Bonus Objectives:

Any further improvements you implement will be considered for bonus recognition. These optional challenges introduce you to techniques used in modern deep learning - and in this campaign, they are your path from group-stage survival to lifting the trophy:

- Cross-Entropy Loss - clinical finishing. MSE spreads its effort everywhere; cross-entropy puts the ball exactly where it must go. The single highest-impact upgrade for classification - implement it and watch your accuracy jump.
- ReLU Activation - fresh legs. tanh tires in deep networks (vanishing gradients); ReLU keeps the gradients running for the full ninety minutes.
- Weight Initialization - pre-season conditioning. A squad that starts the season in the right shape converges faster. Principled initialization sets your parameters up to train smoothly from kickoff.
- Dropout - squad rotation. A team that over-relies on its star players collapses when they're marked out of the game. Randomly resting neurons during training forces the whole squad to learn robust, transferable skills.
- Momentum - counter-attacking pace. Plain gradient descent slows down in midfield. Momentum builds speed in a consistent direction and carries your optimizer through flat, difficult stretches of the loss landscape.